Course: Linear Algebra and Optimization **Course Code: 23MA4BSLAO**

Unit 4: Eigenvalues and Eigenvectors

I. Eigenvalues and Eigenvectors of a linear transformation:

A scalar λ is called an eigenvalue of a linear transformation $T: V \to V$ if there is a non-zero vector X such that $T(X) = \lambda X$. The vector X is called an eigenvector of the linear transformation T corresponding to λ.
The set of all eigenvectors of $\boldsymbol{\lambda}$ along with the zero-vector form subspace called eigenspace of $\boldsymbol{\lambda}$.

Any linear transformation $T: V \to V$ then λ is an eigenvalue of the matrix of the linear transformation, A and X is the corresponding eigenvector.

Theorem: The eigenvalues and eigenvectors of a matrix $A \in M_{n\times n}$ satisfies the following:

i. An eigenvalue of A is a scalar λ such that $|A - \lambda I| = 0$.
ii. The eigenvectors of A corresponding to λ are the nonzero solutions of $(A - \lambda I)x = 0$ i.e. the null space of $A - \lambda I$.
iii. Characteristic polynomial of A is $|A - \lambda I| = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$.

Theorem: Let A be a square matrix. Then the following are equivalent.

a) A scalar λ is an eigen value of A .
b) The matrix $M = A - \lambda I$ is singular.
c) The scalar λ is a root of the characteristic polynomial.

Note:

i. The Characteristic Equation of 2 by 2 matrix: $\lambda^2 - trace(A)\lambda + |A| = 0$.
ii. For 3 by 3 matrix: $-\lambda^3 + trace(A)\lambda^2 - \left(\sum_{i=j} M_{ij}\right)\lambda + |A| = 0$ where M_{ij} is the principal minors of A.

II. Properties of eigenvalues and eigen vectors

1) There are infinitely many eigenvectors corresponding to each eigenvalue but for a given eigenvector the eigenvalue is unique.
2) Eigenvectors of a matrix A corresponding to distinct eigenvalues are linearly independent.
3) Trace of A is equal to the sum of the eigenvalues of A .
4) Determinant of A is equal to product of eigenvalues A .
5) A and A^T has same eigenvalues.
6) Eigenvectors of distinct eigenvalues of real symmetric matrices are orthogonal.
7) If A is a square matrix then the following statements are equivalent:
   a) The eigenvalues are non-zero ($|A| \neq 0$).
   b) A^{-1} exists and $1/\lambda$ is the eigenvalue of A^{-1}
8) Zero is one of the eigenvalue iff A is singular.
9) If λ is an eigenvalue of A then
   a) λ^m is the eigenvalue of A^m .
   b) $k\lambda$ is the eigenvalue of kA .

III. Cayley-Hamilton Theorem:

Statement: Every Square Matrix satisfies its own characteristic equation.

Apply Cayley-Hamilton Theorem to find or Compute using minimal conventional operations.

1) A^{-1} if $A=\begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$.

2) A^4 given $A=\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$.

3) A^{-1} given $A=\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$. Also Compute A^8.

4) A^{-1}, A^{-2}, A^{-3} if $A=\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$.

5) Given $M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ prove that $(4M - 3I)^{-2} = M^{-4}$.

6) Given $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ find M^{-2}.

IV. Eigen Spaces:

Theorem: (The eigenspace of A corresponding to λ)

If $T: V \to V$ is a linear transformation then the subset of all eigenvectors corresponding to λ along with the zero vector is a subspace of V and is called the eigen space of λ denoted by $E(\lambda)$.

Note:

1. If an eigenvalue λ occurs as a multiple root (k times) of the characteristic polynomial, then λ has algebraic multiplicity (A.M.) k.
2. The number of independent eigenvectors corresponding to a given eigenvalue is the geometric multiplicity (G.M.) and hence the dimension of the eigenspace of an eigenvalue λ of A is the geometric multiplicity of λ.
3. $0 < G.M. \leq A.M.$ and if $A.M = 1$ then $G.M. = 1$.
4. The eigenvalue λ is said to be defective if the geometric multiplicity of λ is less than its algebraic multiplicity.
5. An $n \times n$ matrix A is said to be defective if it doesn't have n linearly independent eigenvectors. Equivalently, an $n \times n$ matrix A is said to be defective if it has a defective eigenvalue.

Obtain algebraic and geometric multiplicity of the eigenvalues for the following linear transformations (or Matrices) by finding the eigenspaces and hence identify the defective eigenvalues/matrices.

1) $A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$ $\lambda = 1 + i\sqrt{2}, 1 - i\sqrt{2}.$

2) $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ $\lambda = 1 + i, 1 - i.$

3) $T: R^2 \to R^2$ defined by $T(x, y) = (3x + 3y, x + 5y)$.
4) $T: R^3 \to R^3$ defined by $T(x, y, z) = (x - y + z, x - y + z, x - 2y + 2z)$ Ans: $\lambda = 0,1$
5) $T: P_1 \to P_1$ defined by $T(at + b) = (3a + 5b)t - (2a + 4b)$.
6) $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix}$ $\lambda = 0,0,2$
7) $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ $\lambda = 1,1,3$
8) $T: R^3 \to R^3$ defined by $T(x, y, z) = (2x - 5y, 2y, -x + 4y + z)$. Ans: Defective
9) $T: P_2 \to P_2$ defined by $T(at^2 + bt + c) = (2a - c)t^2 + (2a + b - 2c)t + (-a + 2c)$.
10) $T: R^3 \to R^3$ defined by $T = \begin{bmatrix} 2 & 0 & 0 \\ -1 & -1 & 9 \\ 0 & -1 & 5 \end{bmatrix}$. Ans: Defective
11) $D: V \to V$ defined by $D(f) = \frac{df}{dt}$, where V is the space of functions with basis $S = \{\sin t, \cos t\}$.
12) Let $T: S \to S$ is defined by $T(A) = BA + AB$, where $A \in S$, and $S = \left\{\begin{bmatrix} a & b \\ b & c \end{bmatrix} : a, b, c \in \mathbb{R}\right\}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Given distinct eigen values of T are $\lambda = 0, 2, -2$.
13) $A = \begin{bmatrix} -1 & 1 & 1 \\ -1 & 1 & 1 \\ -2 & 2 & 1 \end{bmatrix}$ Ans: Defective
14) Given $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and eigenvalues of M are $\lambda = 0,2$. Find the algebraic and geometric multiplicity of each λ and hence find eigenspace corresponding to each eigenvalue of $A = M^2 + \frac{1}{2}M$.
15) Let $f(t) = t^3 + 2t^2 + 3$ be the polynomial of degree 3 and X is the eigen vector of A corresponding to eigen value λ. Prove that X is the eigenvector of $f(A)$ corresponding to eigenvalue $f(\lambda)$.

Minimal polynomial and Jordan Canonical Forms:

Monic Polynomial: A polynomial $f(t)$ is monic if its leading coefficient equals one.

Minimal polynomial: Let A be a square matrix. Let $J(A)$ denote the collection of polynomials for which A is a root that is, for which $f(A) = 0$. The set is not empty[because the Cayley-Hamilton theorem tells us that the characteristic polynomial belongs to $J(A)$]. Let $m(t)$ denote the monic polynomial of lowest degree in $J(A)$. We call $m(t)$ the minimal polynomial of the matrix.

- Suppose M is a block diagonal matrix with diagonal blocks $A_1, A_2 ....A_n$. Then the characteristic polynomial of M is the product of the characteristic polynomials of the diagonal blocks A_i.
- Suppose M is a block diagonal matrix with diagonal blocks $A_1, A_2 ....A_n$. Then the minimal polynomial of M is equal to the least common multiple of the minimal polynomials of the diagonal blocks A_i.

1) Find the minimal polynomial of

a) $\begin{bmatrix} 2 & 2 & -5 \\ 3 & 7 & -15 \\ 1 & 2 & -4 \end{bmatrix}$.

c) $\begin{bmatrix} 3 & -2 & 2 \\ 4 & -4 & 6 \\ 2 & -3 & 5 \end{bmatrix}$.

b) $\begin{bmatrix} 4 & -2 & 2 \\ 6 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$.

d) $\begin{bmatrix} 3 & 1 & -1 \\ 2 & 4 & -2 \\ -1 & -1 & 3 \end{bmatrix}$.

2) Find the characteristic and minimal polynomial of following

i) $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 1 & -4 \end{bmatrix}$.

iii) $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$.

ii) $\begin{bmatrix} 2 & 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix}$.

V. **Jordan Canonical Forms**:

1) Find the Jordan canonical forms of T given by characteristic and minimal polynomial of T respectively $f(t)=(t-2)^4(t-5)^3$ and $m(t)=(t-2)^2(t-5)^3$.

2) Write Jordan canonical form or blocks if $f(t)=(t-2)^5$ is characteristic polynomial and $m(t)=(t-2)^2$ is the minimal polynomial.

3) Write Jordan canonical form or blocks if $f(t)=(t-2)^2(t-3)$ is characteristic polynomial and $m(t)=(t-2)(t-3)$ is the minimal polynomial.

4) Write Jordan canonical form or blocks if $f(t)=(t-4)^4$ is characteristic polynomial and $m(t)=(t-4)^2$ is the minimal polynomial.

5) Find the Jordan canonical form of $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & 0 \\ 1 & -2 & 3 \end{bmatrix}$.